

E-COMMERCE SALES & PROFITABILITY ANALYTICS DASHBOARD

Project Overview: This project analyzes e-commerce sales transactions, customer behavior, product performance, profitability, discounts, and returns to identify business growth opportunities and areas of profit leakage. The dashboard was developed using Microsoft Excel with Power Pivot, Data Modeling, DAX Measures, Pivot Tables, Pivot Charts, and Interactive Slicers to simulate a real-world Business Intelligence reporting solution.

Data Model

The solution follows a Star Schema data model consisting of:

Fact Tables

Orders

Returns

Dimension Tables

Customers

Products

Date Table

Relationships

Customers → Orders

Products → Orders

Date Table → Orders

Returns → Orders

Dashboard Statistics

Orders: 10,000

Customers: 1,000

Products: 500

Returns: 788

Measures: 10+

Dashboard Pages: 3

Tools & Techniques Used

Microsoft Excel

Advanced Excel

Pivot Tables

Pivot Charts

Conditional Formatting

Interactive Slicers

Power Pivot

Data Modeling

Relationship Management

Star Schema Design

DAX Measures

Total Sales

Total Profit
Profit Margin %
Orders Count
Average Order Value (AOV)
Return Rate %
Total Customers
Refund Amount

Dashboard Pages

1. Executive Dashboard

Provides a high-level business overview through:

Total Sales
Total Profit
Profit Margin
Total Orders
Customer Count
Average Order Value
Return Rate
Refund Amount

Visualizations include:

Monthly Sales Trend
Monthly Profit Trend
Sales by Category
Profit by Region

2. Customer & Product Analytics

Provides insights into:

Top Customers by Revenue
Top Products by Sales
Product Profitability
Customer Segmentation
Revenue Distribution

Purpose:

Identify key customers and high-performing products.

3. Profitability & Returns Analytics

Provides insights into:

Discount Impact on Profitability
Return Analysis
Refund Analysis
Low-Profit Products
Regional Return Performance

Purpose:

Identify profit leakage and operational improvement opportunities.

Key Findings

Electronics is the highest revenue-generating category.

A small percentage of customers contribute a significant portion of total revenue.

Higher discount levels have a negative impact on profit margins.

Certain products generate strong sales but comparatively lower profits.

Product returns represent a measurable source of revenue loss.
Regional performance varies significantly across sales and profitability metrics.

Business Recommendations

Focus marketing efforts on high-value customer segments.
Review discount policies for low-margin products.
Investigate products with high return frequency.
Improve inventory planning for high-performing categories.
Develop targeted strategies for underperforming regions.
Optimize pricing to improve overall profit margins.

Skills Demonstrated

Power Pivot
Dashboard Design
Business Intelligence Reporting
Data Modeling
DAX Calculations
Dashboard Design
KPI Development
Profitability Analysis
Customer Analytics
Return Analysis
Data Visualization
Executive Reporting

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Advanced Excel |